Two-Way Frequency Tables & Conditional Probability

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Student-Produced Response Questions (SPRs): Foundational Fluency (1–7)

1 [Foundational] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

In a model for the municipal library, in a calculation for the science museum (case 4), if $P(A) = \frac{19}{20}$, what is $P(\text{not } A)$?

Answer: $1/20$. $P(\text{not } A) = 1 - \frac{19}{20} = 1/20$

2 [Foundational] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

In a model for the ceramic studio, a survey of 229 residents found that 51 are from an urban area and commuting by public transit, 31 are from an urban area and not commuting by public transit, 50 are from a rural area and commuting by public transit, and 97 are from a rural area and not commuting by public transit. What is the probability that a randomly selected resident is from an urban area?

Answer: $82/229$. $P(\text{from an urban area}) = \frac{82}{229} = 82/229$

3 [Foundational] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

The two-way table summarizes a survey of visitors at the botanical conservatory classified by membership status and preference between adult tickets and student tickets. If one visitor who is a Member is selected at random, what is the probability that the visitor prefers adult tickets?

Answer: 0. From the two-way table, there are 30 total Members, and 12 of those Members prefer adult tickets. Therefore, the conditional probability is $\frac{12}{30}$.

4 [Foundational] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

The two-way table summarizes a survey of visitors at the municipal library classified by membership status and preference between hardcover books and paperback books. If a visitor who prefers hardcover books is selected at random, what is the probability that the visitor is a Non-Member?

Answer: 3. From the two-way table, the total number of visitors who prefer hardcover books is 30, of whom 16 are Non-Members. Thus the required conditional probability is $\frac{16}{30}$.

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5 [Foundational] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

The two-way table summarizes a survey of visitors at the community theater classified by membership status and preference between small planters and large planters. If one visitor from the entire survey is selected at random, what is the probability that the visitor is a Non-Member who prefers large planters?

Answer: 2. Reading the two-way table, the total number of visitors surveyed is 82, and the cell for Non-Members who prefer large planters contains 29. Thus the probability is $\frac{29}{82}$.

6 [Foundational] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

The two-way table summarizes a survey of visitors at the regional bakery classified by membership status and preference between standard passes and express passes. If one visitor who is a Member is selected at random, what is the probability that the visitor prefers standard passes?

Answer: 1. From the two-way table, there are 39 total Members, and 18 of those Members prefer standard passes. Therefore, the conditional probability is $\frac{18}{39}$.

7 [Foundational] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

The two-way table summarizes a survey of visitors at the coastal aquarium classified by membership status and preference between oak saplings and maple saplings. If a visitor who prefers oak saplings is selected at random, what is the probability that the visitor is a Non-Member?

Answer: 0. From the two-way table, the total number of visitors who prefer oak saplings is 39, of whom 19 are Non-Members. Thus the required conditional probability is $\frac{19}{39}$.

Part I — Student-Produced Response Questions (SPRs): Intermediate Application (8–14)

8 [Intermediate] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

In a model for the meteorological station, in a group of 50 students, 30 are taking Spanish. What is the probability that a randomly chosen student is NOT taking Spanish?

Answer: 0.40. Not taking Spanish = $50 - 30 = 20$. Probability = $20/50 = 0.40$.

9 [Intermediate] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

In a model for the urban bike-share station, in a calculation for the public transit depot (case 32), if $P(E) = 0.82$, what is $P(E')$?

Answer: 0.18. By the complement rule of probability, $P(E') = 1 - P(E) = 1 - 0.82 = 0.18$.

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10 [Intermediate] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

In a model for the sports complex, a student has 4 blue pens and 1 red pen. If one pen is picked at random, what is the probability it is red?

Answer: 0.20. Total = 5. Probability of red = $1/5 = 0.20$.

11 [Intermediate] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

In a model for the wind energy cooperative, a two-way table of $N = 102$ observations on science track records cell counts $p_1 = 34$ for (Section A, Physics) and $q_2 = 24$ for (Section B, Biology). What is the diagonal sum $t = p_1 + q_2$?

Answer: 58. Summing the two frequencies gives $34 + 24 = 58$.

12 [Intermediate] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

In a model for the astronomical society, a survey of 210 residents found that 130 are from an urban area and 80 are from a rural area. Of those who are from an urban area, 114 are also commuting by public transit. Of those who are from a rural area, 58 are also commuting by public transit. How many residents in total are commuting by public transit?

Answer: 172. Total who are commuting by public transit $= 114 + 58 = 172$

13 [Intermediate] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

In a model for the film archive, a survey of 99 patients found that 31 are receiving the treatment and 68 are receiving the placebo. Of those who are receiving the treatment, 24 are also showing improvement. Of those who are receiving the placebo, 20 are also showing improvement. How many patients in total are showing improvement?

Answer: 44. Total who are showing improvement $= 24 + 20 = 44$

14 [Intermediate] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

In a model for the mountain observatory, two events A and B are independent. If $P(A) = 0.4$ and $P(B) = 0.7$, what is $P(A \text{ and } B)$?

Answer: 0.28. For independent events, $P(A \cap B) = P(A) \cdot P(B) = 0.4 \times 0.7 = 0.28$.

Part I — Student-Produced Response Questions (SPRs): Challenging Synthesis (15–20)

15 [Challenging] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

In a model for the maple syrup farm, if $P(A) = 0.4$, $P(B) = 0.5$, and A and B are independent, what is $P(A \text{ or } B)$?

Answer: 0. $P(A \cap B) = 0.4 \times 0.5 = 0.20$. $P(A \cup B) = 0.4 + 0.5 - 0.20 = 0.70$.

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16 [Challenging] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

In a model for the planetarium, a bag has 4 red, 6 blue, and 2 green marbles. One marble is drawn, observed, and REPLACED, then a second is drawn. What is the probability that both are blue?

Answer: 0. Total = 12. With replacement: $\frac{6}{12} \times \frac{6}{12} = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$.

17 [Challenging] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

In a model for the regional airport shuttle, a test consists of 3 true/false questions. If a student guesses on all 3, what is the probability of getting all 3 correct?

Answer: 1. The probability of guessing correctly on each question is $1/2$. For 3 independent questions, the probability is $(1/2)^3 = 1/8 = 0.125$.

18 [Challenging] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

In a model for the wind energy cooperative, if $P(A \mid B) = 0.6$ and $P(B) = 0.5$, what is $P(A \text{ and } B)$?

Answer: 2. $P(A \cap B) = P(B) \cdot P(A \mid B) = 0.5 \times 0.6 = 0.30$.

19 [Challenging] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

In a model for the marine research vessel, two standard dice are rolled. What is the probability that the sum of the numbers is 7?

Answer: 1. Pairs summing to 7: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1) — 6 outcomes out of 36: $6/36 = 1/6$.

20 [Challenging] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

At the botanical conservatory, a group of 30 workshop participants includes 14 who selected adult tickets and 16 who selected student tickets. If one participant from this group is selected at random, what is the probability that the participant selected adult tickets?

Answer: 0. There are 14 participants who selected adult tickets out of 30 total participants, so the probability is $\frac{14}{30}$.

Part II — Multiple-Choice Questions (MCQs): Foundational Fluency (21–28)

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21 [Foundational] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

The two-way table summarizes a survey of visitors at the mountain observatory classified by membership status and preference between glass jars and ceramic mugs. If one visitor from the entire survey is selected at random, what is the probability that the visitor is a Non-Member who prefers ceramic mugs?

a. $\frac{35}{55}$

b. $\frac{35}{58}$

c. $\frac{23}{100}$

d. $\frac{35}{100}$ ✓

Reading the two-way table, the total number of visitors surveyed is 100, and the cell for Non-Members who prefer ceramic mugs contains 35. Thus the probability is $\frac{35}{100}$.

22 [Foundational] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

The two-way table summarizes a survey of visitors at the science museum classified by membership status and preference between morning shuttles and afternoon shuttles. If one visitor who is a Member is selected at random, what is the probability that the visitor prefers morning shuttles?

a. $\frac{24}{106}$

b. $\frac{48}{106}$

c. $\frac{24}{48}$ ✓

d. $\frac{24}{45}$

From the two-way table, there are 48 total Members, and 24 of those Members prefer morning shuttles. Therefore, the conditional probability is $\frac{24}{48}$.

23 [Foundational] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

The two-way table summarizes a survey of visitors at the urban bike-share station classified by membership status and preference between solar panels and battery modules. If a visitor who prefers solar panels is selected at random, what is the probability that the visitor is a Non-Member?

a. $\frac{26}{48}$

b. $\frac{22}{48}$ ✓

c. $\frac{22}{61}$

d. $\frac{22}{112}$

From the two-way table, the total number of visitors who prefer solar panels is 48, of whom 22 are Non-Members. Thus the required conditional probability is $\frac{22}{48}$.

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24 [Foundational] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

The two-way table summarizes a survey of visitors at the solar microgrid facility classified by membership status and preference between cotton shirts and linen shirts. If one visitor from the entire survey is selected at random, what is the probability that the visitor is a Non-Member who prefers linen shirts?

a. $\frac{41}{118}$ ✓

b. $\frac{41}{64}$

c. $\frac{41}{67}$

d. $\frac{26}{118}$

Reading the two-way table, the total number of visitors surveyed is 118, and the cell for Non-Members who prefer linen shirts contains 41. Thus the probability is $\frac{41}{118}$.

25 [Foundational] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

The two-way table summarizes a survey of visitors at the ceramic studio classified by membership status and preference between 2-pound bags and 5-pound bags. If one visitor who is a Member is selected at random, what is the probability that the visitor prefers 2-pound bags?

a. $\frac{30}{54}$

b. $\frac{30}{124}$

c. $\frac{57}{124}$

d. $\frac{30}{57}$ ✓

From the two-way table, there are 57 total Members, and 30 of those Members prefer 2-pound bags. Therefore, the conditional probability is $\frac{30}{57}$.

26 [Foundational] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

The two-way table summarizes a survey of visitors at the agricultural cooperative classified by membership status and preference between bench seats and floor seats. If a visitor who prefers bench seats is selected at random, what is the probability that the visitor is a Non-Member?

a. $\frac{25}{130}$

b. $\frac{32}{57}$

c. $\frac{25}{57}$ ✓

d. $\frac{25}{70}$

From the two-way table, the total number of visitors who prefer bench seats is 57, of whom 25 are Non-Members. Thus the required conditional probability is $\frac{25}{57}$.

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27 [Foundational] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

The two-way table summarizes a survey of visitors at the publishing house classified by membership status and preference between weekday rentals and weekend rentals. If one visitor from the entire survey is selected at random, what is the probability that the visitor is a Non-Member who prefers weekend rentals?

a. $\frac{29}{136}$

b. $\frac{47}{136}$ ✓

c. $\frac{47}{73}$

d. $\frac{47}{76}$

Reading the two-way table, the total number of visitors surveyed is 136, and the cell for Non-Members who prefer weekend rentals contains 47. Thus the probability is $\frac{47}{136}$.

28 [Foundational] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

The two-way table summarizes a survey of visitors at the public transit depot classified by membership status and preference between blueberry muffins and cinnamon scones. If one visitor who is a Member is selected at random, what is the probability that the visitor prefers blueberry muffins?

a. $\frac{36}{66}$ ✓

b. $\frac{36}{63}$

c. $\frac{36}{142}$

d. $\frac{66}{142}$

From the two-way table, there are 66 total Members, and 36 of those Members prefer blueberry muffins. Therefore, the conditional probability is $\frac{36}{66}$.

Part II — Multiple-Choice Questions (MCQs): Intermediate Application (29–37)

29 [Intermediate] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

In a model for the art gallery, a standard 6-sided die is rolled. What is the probability of rolling a number greater than 4?

a. $\frac{1}{6}$

b. $\frac{1}{3}$ ✓

c. $\frac{1}{2}$

d. $\frac{2}{3}$

The numbers greater than 4 are 5 and 6 (2 outcomes out of 6): $2/6 = 1/3$.

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30 [Intermediate] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

In a model for the robotics club, a bag contains 5 red, 4 blue, and 6 green marbles. If one marble is drawn at random, what is the probability that it is blue?

a. $\frac{4}{15}$ ✓ **b.** $\frac{1}{3}$

c. $\frac{2}{5}$ **d.** $\frac{1}{4}$

Total marbles = $5 + 4 + 6 = 15$. Probability of blue = $4/15$.

31 [Intermediate] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

In a model for the culinary institute, if the probability of rain tomorrow is 0.35, what is the probability that it will NOT rain tomorrow?

a. 0.35 **b.** 0.55

c. 0.65 ✓ **d.** 0.75

Complement rule: $P(\text{not rain}) = 1 - 0.35 = 0.65$.

32 [Intermediate] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

In a model for the water treatment plant, a standard deck has 52 cards. What is the probability of drawing an Ace at random?

a. $\frac{1}{52}$ **b.** $\frac{1}{26}$

c. $\frac{1}{13}$ ✓ **d.** $\frac{1}{4}$

There are 4 Aces in 52 cards: $4/52 = 1/13$.

33 [Intermediate] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

In a model for the sports complex, a spinner has 8 equal sections numbered 1 through 8. What is the probability of spinning an even number?

a. $\frac{1}{4}$ **b.** $\frac{3}{8}$

c. $\frac{1}{2}$ ✓ **d.** $\frac{5}{8}$

Even numbers are 2, 4, 6, 8 (4 out of 8): $4/8 = 1/2$.

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34 [Intermediate] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

In a model for the maple syrup farm, a jar contains 12 red candies and 18 green candies. What is the probability of randomly choosing a red candy?

a. $\frac{2}{5}$ ✓ **b.** $\frac{3}{5}$

c. $\frac{1}{2}$ **d.** $\frac{2}{3}$

Total = $12 + 18 = 30$. Probability = $12/30 = 2/5$.

35 [Intermediate] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

In a model for the planetarium, a weather station predicts that the probability of snow tomorrow is $\frac{3}{8}$. What is the probability that it will NOT snow tomorrow?

a. $\frac{1}{8}$ **b.** $\frac{3}{8}$

c. $\frac{5}{8}$ ✓ **d.** $\frac{7}{8}$

$1 - \frac{3}{8} = \frac{5}{8}$.

36 [Intermediate] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

In a model for the regional airport shuttle, a box contains 20 pens: 8 black, 7 blue, and 5 red. One pen is picked at random. What is the probability that it is NOT red?

a. $\frac{1}{4}$ **b.** $\frac{3}{4}$ ✓

c. $\frac{2}{5}$ **d.** $\frac{3}{5}$

Non-red pens = $8 + 7 = 15$. Probability = $15/20 = 3/4$.

37 [Intermediate] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

In a model for the wind energy cooperative, in a calculation for the community orchard (case 19), a coin is flipped once. What is the probability of landing on tails?

a. 0 **b.** 0.25

c. 0.50 ✓ **d.** 1.0

A fair coin has 2 outcomes: $1/2 = 0.50$.

Part II — Multiple-Choice Questions (MCQs): Challenging Synthesis (38–45)

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38 [Challenging] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

At the municipal library, a group of 33 workshop participants includes 16 who selected hardcover books and 17 who selected paperback books. If one participant from this group is selected at random, what is the probability that the participant selected hardcover books?

a. $\frac{17}{33}$

b. $\frac{16}{17}$

c. $\frac{17}{16}$

d. $\frac{16}{33}$ ✓

There are 16 participants who selected hardcover books out of 33 total participants, so the probability is $\frac{16}{33}$.

39 [Challenging] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

At the community theater, a group of 36 workshop participants includes 18 who selected small planters and 18 who selected large planters. If one participant from this group is selected at random, what is the probability that the participant selected small planters?

a. 1839

b. 1842

c. $\frac{18}{36}$ ✓

d. $\frac{18}{18}$

There are 18 participants who selected small planters out of 36 total participants, so the probability is $\frac{18}{36}$.

40 [Challenging] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

At the regional bakery, a group of 39 workshop participants includes 20 who selected standard passes and 19 who selected express passes. If one participant from this group is selected at random, what is the probability that the participant selected standard passes?

a. $\frac{19}{20}$

b. $\frac{20}{39}$ ✓

c. $\frac{19}{39}$

d. $\frac{20}{19}$

There are 20 participants who selected standard passes out of 39 total participants, so the probability is $\frac{20}{39}$.

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41 [Challenging] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

At the coastal aquarium, a group of 42 workshop participants includes 22 who selected oak saplings and 20 who selected maple saplings. If one participant from this group is selected at random, what is the probability that the participant selected oak saplings?

a. $\frac{22}{42}$ ✓

b. $\frac{20}{42}$

c. $\frac{22}{20}$

d. $\frac{20}{22}$

There are 22 participants who selected oak saplings out of 42 total participants, so the probability is $\frac{22}{42}$.

42 [Challenging] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

At the mountain observatory, a group of 45 workshop participants includes 24 who selected glass jars and 21 who selected ceramic mugs. If one participant from this group is selected at random, what is the probability that the participant selected glass jars?

a. $\frac{21}{45}$

b. $\frac{24}{21}$

c. $\frac{21}{24}$

d. $\frac{24}{45}$ ✓

There are 24 participants who selected glass jars out of 45 total participants, so the probability is $\frac{24}{45}$.

43 [Challenging] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

At the science museum, a group of 48 workshop participants includes 26 who selected morning shuttles and 22 who selected afternoon shuttles. If one participant from this group is selected at random, what is the probability that the participant selected morning shuttles?

a. $\frac{26}{22}$

b. $\frac{22}{26}$

c. $\frac{26}{48}$ ✓

d. $\frac{22}{48}$

There are 26 participants who selected morning shuttles out of 48 total participants, so the probability is $\frac{26}{48}$.

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44 [Challenging] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

At the urban bike-share station, a group of 51 workshop participants includes 28 who selected solar panels and 23 who selected battery modules. If one participant from this group is selected at random, what is the probability that the participant selected solar panels?

a. $\frac{23}{28}$

b. $\frac{28}{51}$ ✓

c. $\frac{23}{51}$

d. $\frac{28}{23}$

There are 28 participants who selected solar panels out of 51 total participants, so the probability is $\frac{28}{51}$.

45 [Challenging] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

At the solar microgrid facility, a group of 54 workshop participants includes 30 who selected cotton shirts and 24 who selected linen shirts. If one participant from this group is selected at random, what is the probability that the participant selected cotton shirts?

a. $\frac{30}{54}$ ✓

b. $\frac{24}{54}$

c. $\frac{30}{24}$

d. $\frac{24}{30}$

There are 30 participants who selected cotton shirts out of 54 total participants, so the probability is $\frac{30}{54}$.

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